SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – SIH26197**
- **Problem Statement Title-** Student Innovation—Ideas that showcase the rich cultural heritage and traditions of India
- **Theme-** Heritage & Culture
- **PS Category-** Software
- **Team ID-** 57385
- **Team Name (Registered on portal)** Team VirasatX

Team VirasatX

IDEA TITLE: VIRASATX

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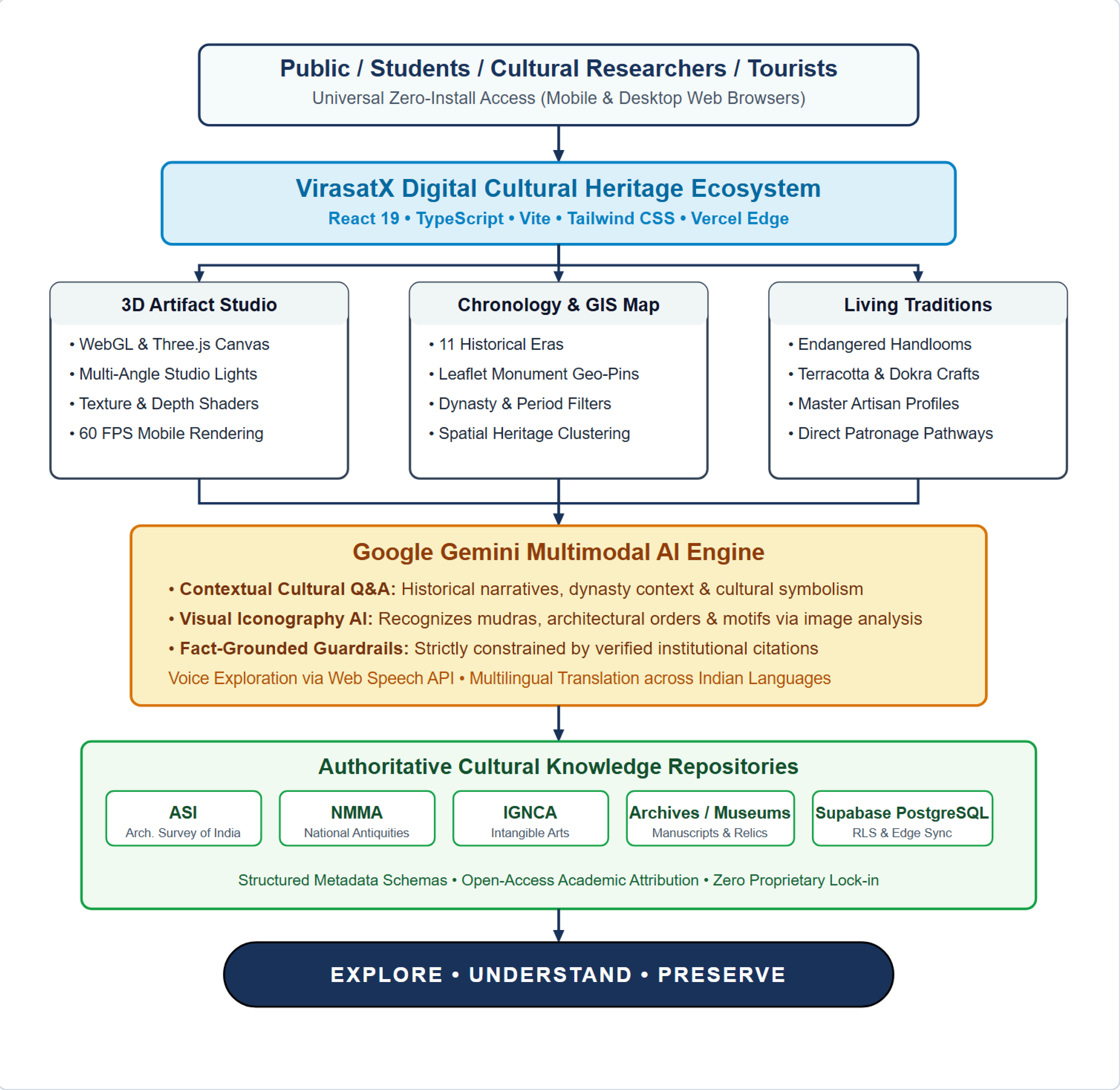
❖Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation of the proposed solution**
  - **Unified Cultural Ecosystem:** VirasatX connects monuments, 3D artifacts, living traditions, timelines, and regional geography into one cohesive, discoverable experience.
  - **Multi-Modal Exploration:** Interactive Three.js WebGL 3D artifact studio, 11-period chronological timeline, Leaflet GIS heritage mapping, and living artisan craft profiles.
  - **Google Gemini Multimodal AI:** Contextual conversational cultural Q&A and visual iconography AI analyzing motifs, mudras, and architectural orders with fact-grounded guardrails.
- **How it addresses the problem**
  - **Synthesizes Fragmented Archives:** Overcomes isolated, disconnected heritage archives by linking assets across historical epochs and geographic regions into an intuitive discovery graph.
  - **Bridges Cultural Context Gap:** Transitions users from passive browsing to deep civilizational context (Dynasty → Period → Artisan Technique → Architectural Symbolism).
  - **Elevates Endangered Traditions:** Provides dedicated digital visibility and direct patronage pathways for endangered craft forms, folklore, and master artisans.
- **Innovation and uniqueness of the solution**
  - **Context-First Philosophy:** "VirasatX connects the cultural context around heritage — not just isolated artifacts or 3D models."
  - **Interconnected Discovery Chain:** Artifact → Story → Period → Place → Tradition → Community → Active Learning & Patronage.



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TECHNICAL APPROACH

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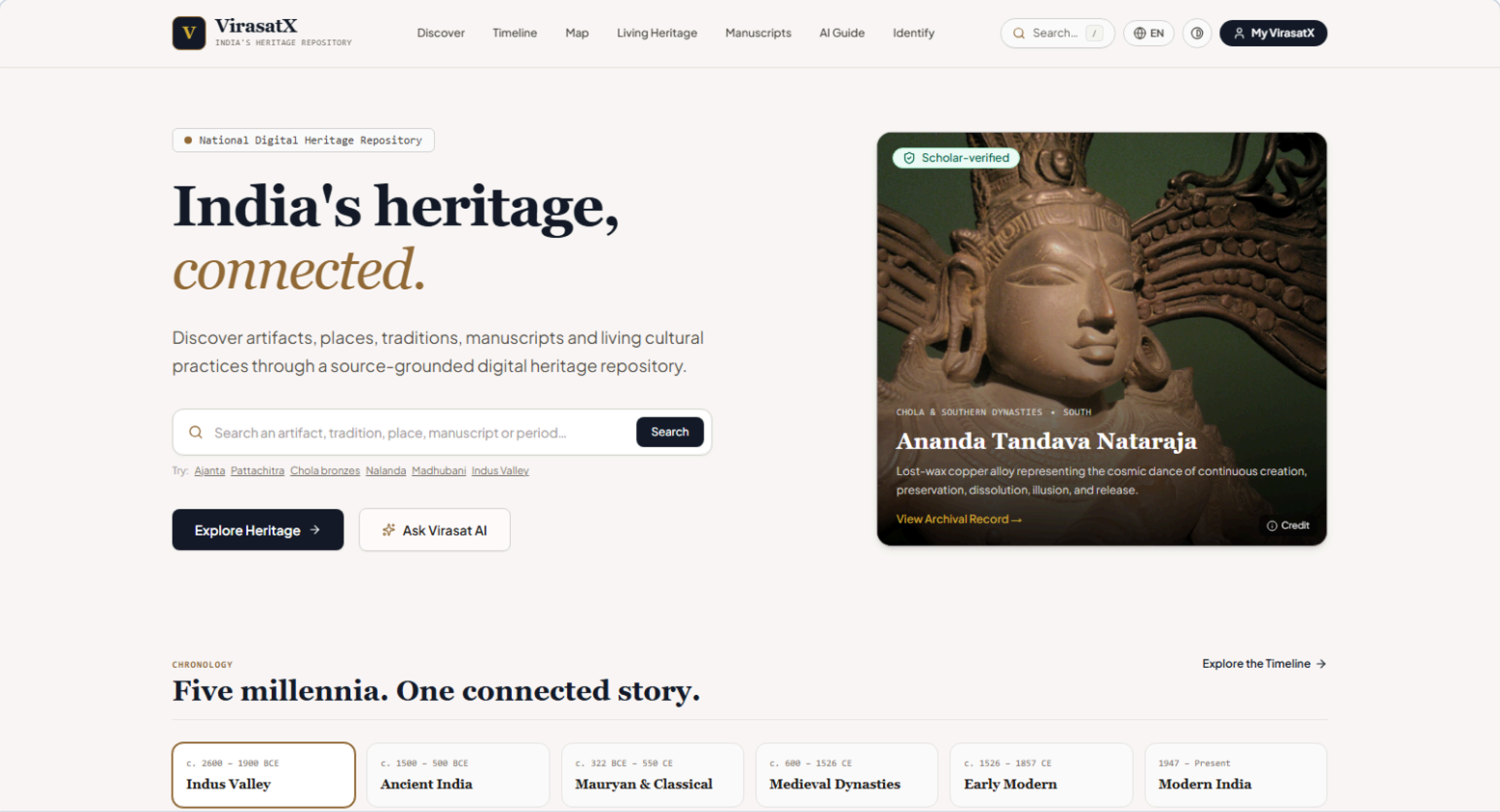
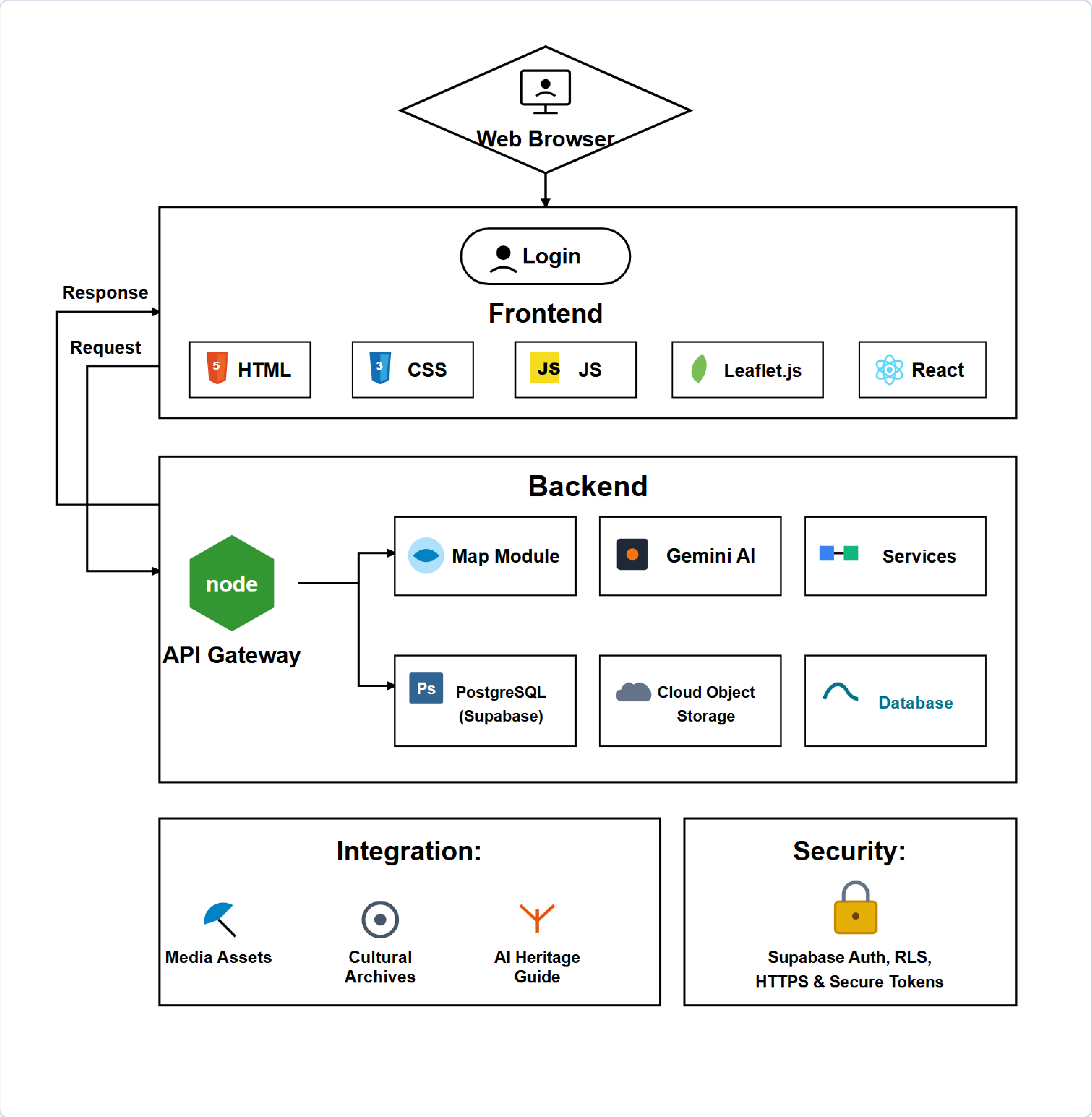
- **Technologies to be used (e.g. programming languages, frameworks, hardware)**
  - **Frontend Framework:** React 19, TypeScript, Vite & Tailwind CSS with Motion for high-fidelity animations.
  - **Multimodal AI Engine:** Google Gemini AI for contextual cultural exploration and visual iconography analysis.
  - **3D Artifact Studio:** Three.js & WebGL for real-time interactive 3D model rendering directly in mobile browsers.
  - **GIS Spatial Mapping:** Leaflet & OpenStreetMap for geographic heritage discovery and cluster exploration.
  - **Voice Interaction:** Web Speech API for voice-assisted queries and hands-free audio guidance.
  - **Backend & Database:** Supabase with PostgreSQL relational database and Row Level Security (RLS).
  - **Cloud & Edge:** Vercel serverless edge deployment ensuring sub-second global response times.
- **Methodology and process for implementation (Flow Charts/Images/ working prototype)**



Live Working Prototype: React 19 • Three.js WebGL • Gemini AI (https://virasatxai.vercel.app/)



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FEASIBILITY AND VIABILITY

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• Analysis of the feasibility of the idea

1. Technical Feasibility:

- **Lightweight Web Architecture:** React 19, TypeScript, and WebGL deliver responsive 3D artifact rendering directly in mobile browsers without demanding native app downloads.
- **Modular Multimodal AI:** Google Gemini multimodal AI orchestrated through serverless edge functions for rapid, low-latency contextual dialogue and iconography assistance.
- **Standardized Cultural Data Schemas:** Structured PostgreSQL & JSON schemas compatible with national heritage classification and archival cataloging conventions.

2. Operational Feasibility:

- **Public Open Datasets:** Built upon documented public domain archives, monument registries, and academic resources without proprietary data dependencies.
- **Zero Infrastructure Overhead:** Serverless deployment on Vercel and Supabase eliminates physical server management and ongoing operational overhead.
- **Modular Content Extensibility:** Structured content taxonomy allows adding new regional monuments, epochs, and craft traditions without platform refactoring.

3. Legal & Ethical Feasibility:

- **Privacy & Data Governance:** Strictly compliant with India's Digital Personal Data Protection (DPDP) Act with zero unauthorized user tracking or commercial data harvesting.
- **Cultural Sensitivity & Attribution:** Strict prompt guardrails enforce respectful, non-distorted historical interpretation with explicit source citations.

• Potential challenges and risks

- **Data Fragmentation:** Heritage information scattered across divergent regional archives with inconsistent naming conventions and periodization.
- **AI Hallucination Risk:** Risk of large language models generating inaccurate historical dates, dynasties, or architectural attributions.
- **Connectivity Variations:** Rich 3D models and high-resolution imagery may experience latency on 3G/4G rural networks.

• Strategies for overcoming these challenges

- **Unified Taxonomy Schemas:** Standardized normalizers reconcile regional spelling variants, dynasties, and historical eras into a single linked graph.
- **Fact-Grounded Prompting:** AI Guide strictly constrained to verified facts with explicit archival citations and transparent uncertainty disclaimers.
- **Progressive Asset Optimization:** WebP compression, low-polygon 3D meshes, and intelligent browser caching ensure sub-second page loads.

❖ Economic Viability & Scalability

- **Cost-Effective Foundation:** Built on modern open-source web technologies with near-zero baseline hosting expenses.
- **Elastic Cloud Scaling:** Serverless edge hosting through Vercel and Supabase scales dynamically on demand from pilot districts to national scale.

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IMPACT AND BENEFITS

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• Potential impact on the target audience

1. Students & Youth:

- Replaces dry textbook memorization with interactive 3D artifact exploration, timeline epoch navigation, and conversational AI guidance.
- Fosters civilizational pride and active engagement among younger generations through visual and interactive learning pathways.

2. Cultural Researchers & Educators:

- Provides a centralized digital platform linking monuments, artifacts, regional periods, and living traditions with institutional citations.
- Accelerates cultural studies with structured epoch relationships, iconography references, and comparative timelines.

3. Traditional Artisans & Craft Communities:

- Creates direct digital visibility for indigenous handlooms, terracotta, metalcasting, and regional craftsmanship.
- Connects traditional craftspeople to cultural discovery pathways, fostering greater public awareness and patronage.

4. Cultural Tourists & Heritage Travelers:

- Promotes context-first responsible travel, encouraging deeper respect and cultural literacy before visiting physical heritage sites.
- Uncovers hidden, lesser-known regional heritage treasures beyond mainstream crowded tourist landmarks.

• Benefits of the solution (social, economic, environmental, etc.)

1. Social & Cultural Benefits:

- Digitally preserves India’s tangible and intangible heritage for future generations.
- Fosters intercultural dialogue and civilizational awareness across diverse Indian regions and languages.

2. Educational & Research Benefits:

- Delivers an open-access cultural learning repository for schools, universities, and self-directed learners nationwide.
- Encourages interdisciplinary exploration connecting architecture, epigraphy, mythology, and craft sciences.

3. Economic & Community Benefits:

- Stimulates regional cultural tourism, dispersing economic benefits to rural communities and artisan clusters.
- Increases awareness and demand for authentic traditional craftsmanship and community-made goods.

4. Digital Preservation & Scalability:

- Creates a future-ready, resilient digital architecture that scales seamlessly across hundreds of regional sites.
- Safeguards vulnerable cultural knowledge against permanent loss through structured digital documentation.

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RESEARCH AND REFERENCES

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• **Details / Links of the reference and research work**

1. Project & Prototype:

VirasatX — India's Heritage Repository

- Live Prototype: https://virasatxai.vercel.app/
- GitHub Repository: https://github.com/ShrilCarpenter/VirasatX
- Core Modules: 3D WebGL Studio, 11-Era Chronology Timeline, Leaflet GIS Map, Gemini AI Guide

2. Institutional Reference Sources:

Institutional sources consulted for taxonomy, cataloging schemas, and heritage documentation:

- Ministry of Culture (https://www.indiaculture.gov.in/)
- Archaeological Survey of India (ASI) (https://asi.nic.in/)
- National Archives of India (https://nationalarchives.nic.in/)
- National Mission on Monuments and Antiquities (NMMA) (http://nmma.nic.in/)
- Museums of India (https://museumsofindia.gov.in/)
- Indira Gandhi National Centre for the Arts (IGNCA) (https://ignca.gov.in/)
- Vedic Heritage Portal (https://vedicheritage.gov.in/)
- UNESCO World Heritage Centre (https://whc.unesco.org/)

3. Technical Standards & Frameworks:

- **W3C WebGL & WebXR Guidelines:** Mobile-optimized interactive 3D rendering standards.
- **CIDOC-CRM & Dublin Core:** Global cultural heritage cataloging and semantic metadata models.
- **Digital Personal Data Protection (DPDP) Act 2023:** Full regulatory compliance for citizen privacy.
- **National Education Policy (NEP 2020):** Integration with Indian Knowledge Systems (IKS) pedagogy.

VIRASATX

"From scattered heritage knowledge to one connected cultural experience."

EXPLORE • UNDERSTAND • PRESERVE

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